

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B Tech CSE

PHASE – 2 (Design)

Group No : D7

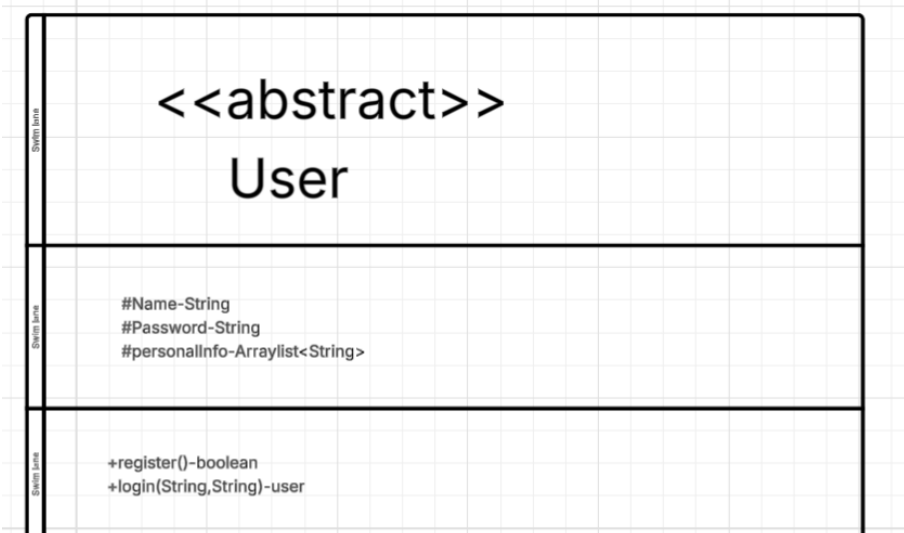
Team Members :

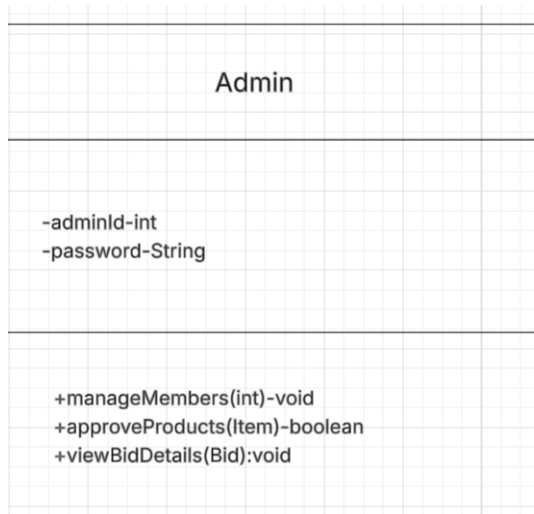
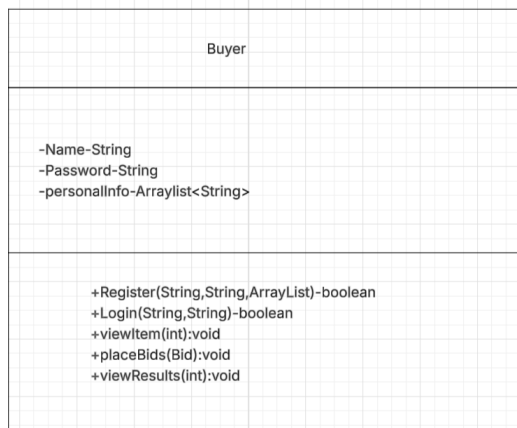
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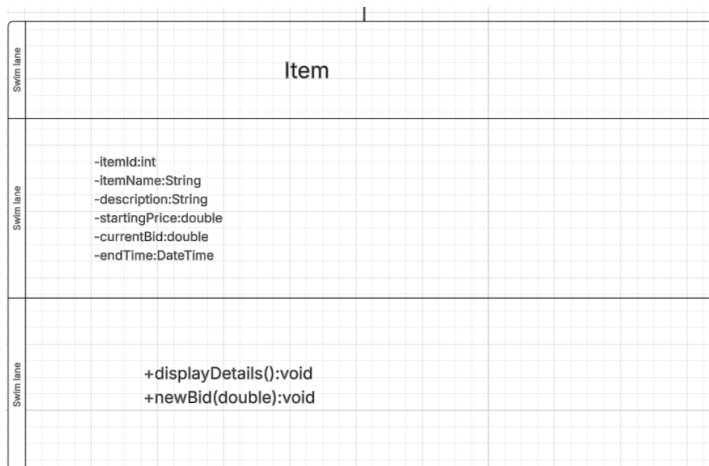
1. Problem Title: Online auction system

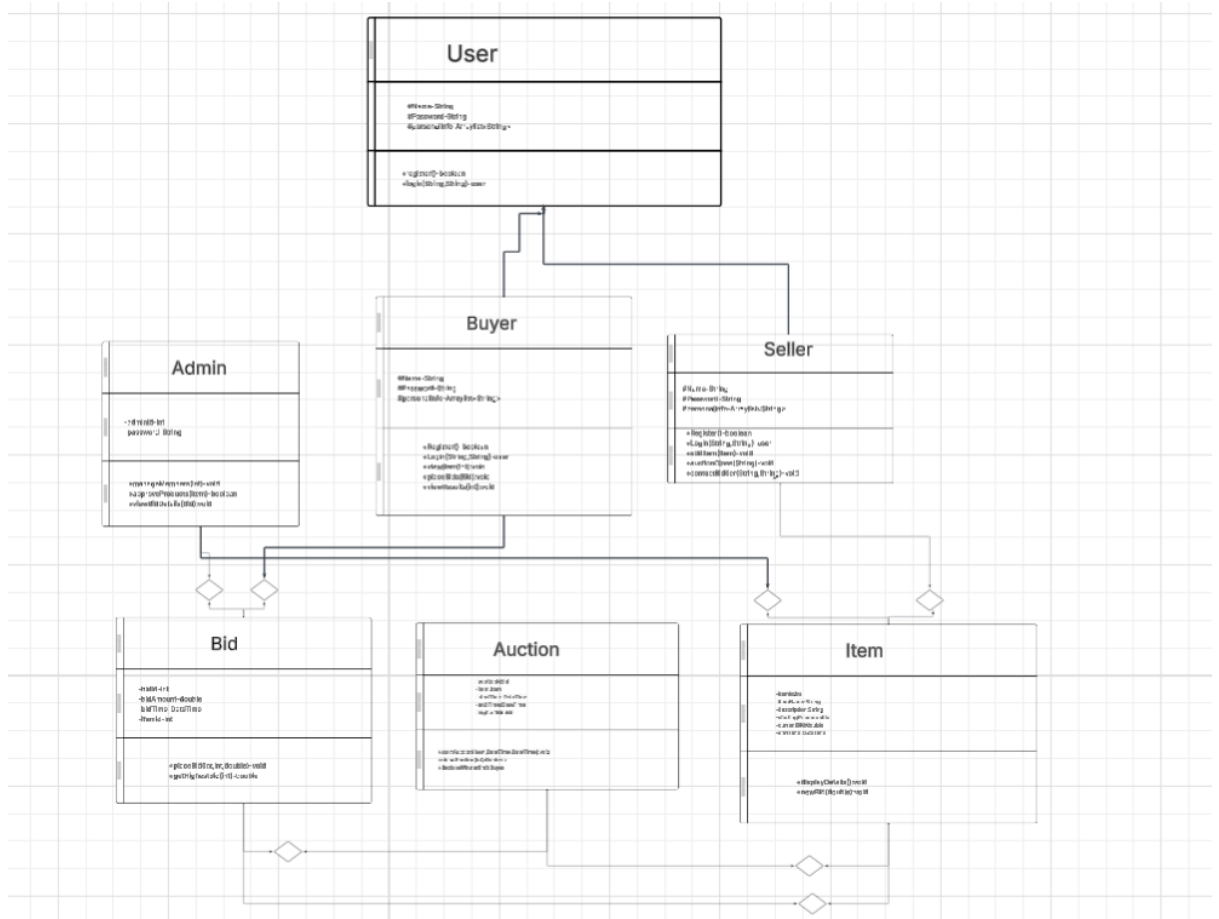
STRUCTURAL DESIGN

2. Class diagram (draw class diagram that shows different classes and their relationships. Use UML notations)









BEHAVIORAL DESIGN

3. Use-case diagram

a. List of actors

1. Seller

2. Buyer

3. Admin

b. List of use-cases

1. Describe each use case here in 1 or 2 sentences.

Common Use Cases

1. Register

Allows a new user (buyer or seller) to create an account in the system by providing required details.

2. Login

Allows registered users to access the system securely using their credentials.

Seller Use Cases

3. Add Item

Allows the seller to add a new product with details like name, description, and starting price for auction.

4. Close Auction

Allows the seller to close the auction after the end time and stop further bidding.

5. Contact Bidder

Allows the seller to communicate with the highest bidder after the auction ends.

Buyer Use Cases

6. View Item

Allows the buyer to browse and view available items along with their details.

7. Place Bids

Allows the buyer to place bids on items during the auction period.

8. View Results

Allows the buyer to check the auction outcome and see the winning bid.

Admin Use Cases

9. Manage Members

Allows the admin to manage user accounts, including adding, updating, or removing users.

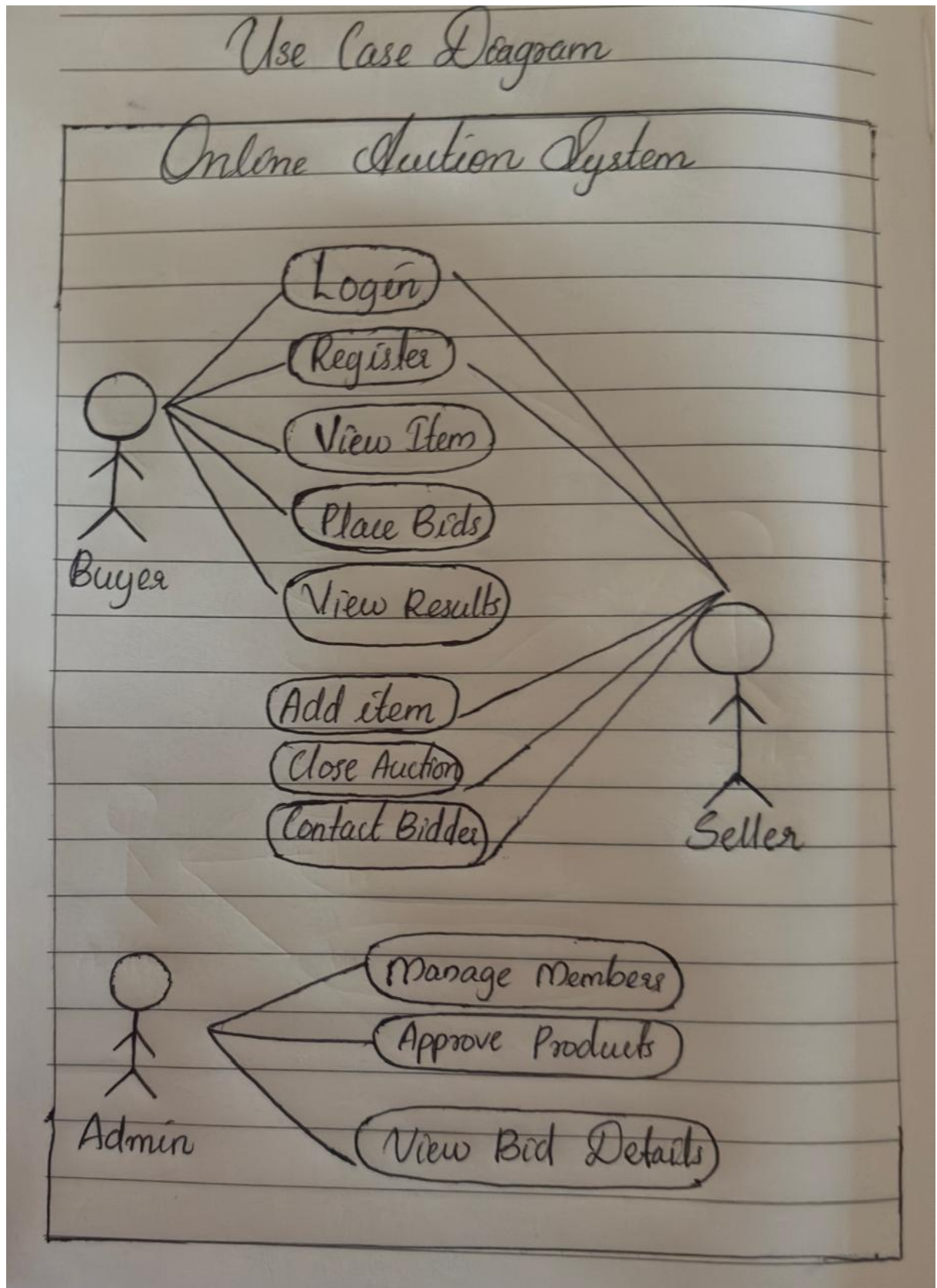
10. Approve Products

Allows the admin to review and approve items before they are listed for auction.

11. View Bid Details

Allows the admin to monitor and view all bidding activities in the system.

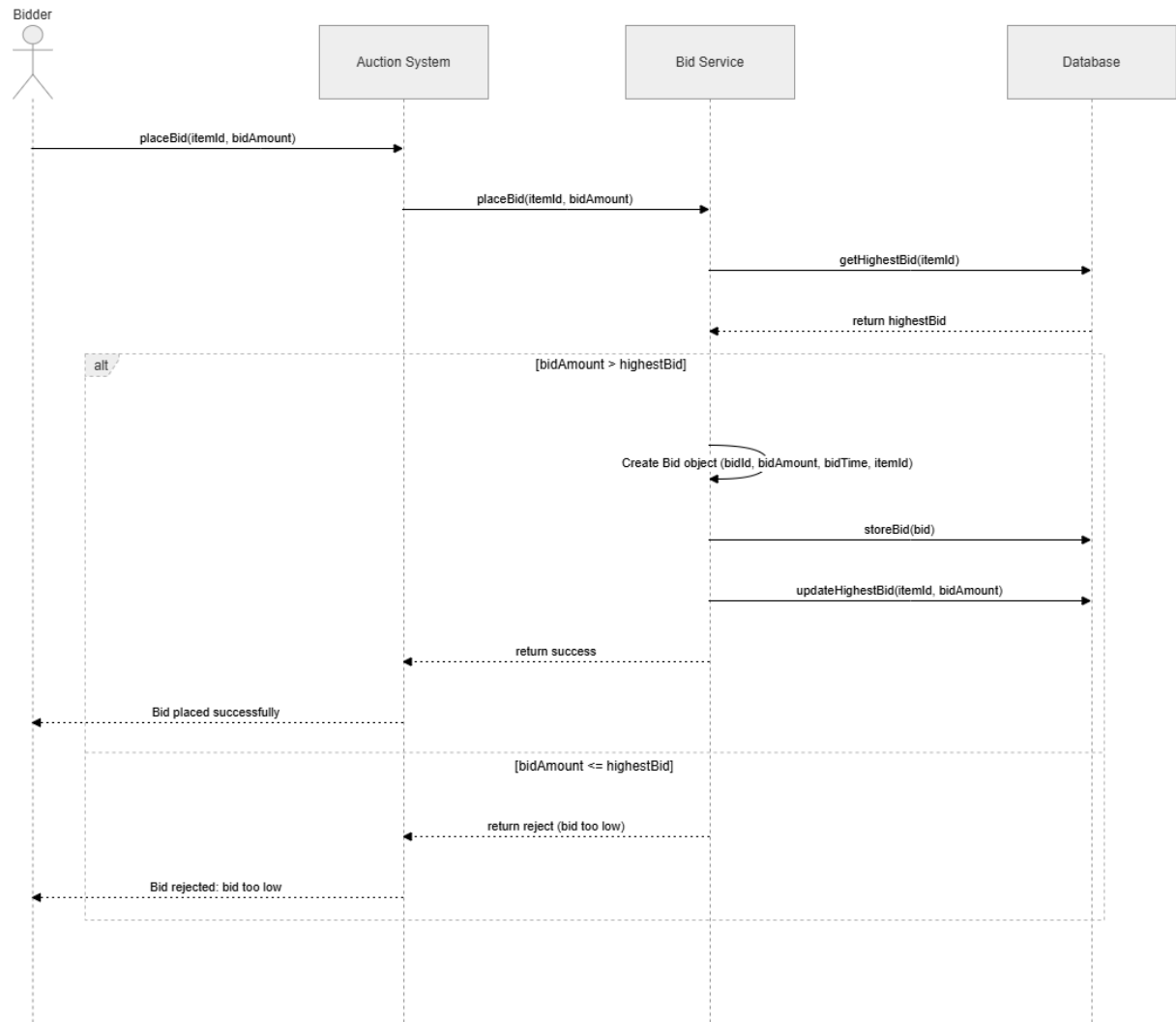
c. Use case diagram (use UML notations)



4. Sequence diagram (use UML notations)

-Sequence diagram for placing a bid.

The scenario shows a **user trying to place a bid in an online auction system**. The system checks the current highest bid before accepting it. Since the user's bid amount is **less than or equal to the highest bid**, the system rejects the request. Finally, a message is sent back to the user indicating that the **bid was too low and not accepted**.



-Sequence diagram of closing bid

When the auction time expires or the seller manually closes it, if there's a top bidder, the system officially selects them as the winner and sends a notification to both the seller and the winning bidder. Else, if the auction finishes without a single bid, the system simply closes the bidding, and the item remains unsold.

